ORC Station Operator Guide

Indonesia ORC Deployment — Spring 2026

American Red Cross / Palang Merah Indonesia

Version 1.0 — April 2026 | Audience: PMI field staff | Built 2026-04-09

Contents

ORC Station Operator Guide **2**
Contents . . . 2
What the Station Does . . . 2
1. Status LED — Is the Station OK? . . . 2
2. Normal Operation . . . 3
3. Checking the Station Remotely . . . 3
4. Power Button . . . 4
5. Power Cycle — How the Station Wakes and Sleeps . . . 4
6. Maintenance Mode . . . 5
7. Routine Maintenance . . . 6
8. Troubleshooting — Quick Fixes . . . 7
9. What NOT to Do . . . 8
10. Sensor Reference . . . 8
11. Extending the Station — Available Relay Channels . . . 12
12. Emergency Contacts . . . 15

ORC Station Operator Guide

For: PMI field staff responsible for monitoring and maintaining ORC stations **Sites:** Sukabumi (solar) and Jakarta (AC power) **Version:** 1.0 — April 2026

Contents

- What the Station Does
- 1. Status LED
- 2. Normal Operation
- 3. Checking the Station Remotely
- 4. Power Button
- 5. Power Cycle — Wake and Sleep
- 6. Maintenance Mode
- 7. Routine Maintenance
- 8. Troubleshooting — Quick Fixes
- 9. What NOT to Do
- 10. Sensor Reference
- 11. Extending the Station
- 12. Emergency Contacts

What the Station Does

The station automatically records short videos of the river every 15 minutes, measures water velocity from the video, and calculates river flow (discharge). Results upload to a central server over LTE cellular data.

You do not need to operate the station day-to-day. It runs unattended. This guide covers what to check, how to tell if something is wrong, and what to do about it.

---

1. Status LED — Is the Station OK?

The LED is visible through the enclosure window without opening the box.

LED Color	Pattern	Meaning	Action
Green	Steady	Healthy, waiting for next capture	None — normal
Green	Flashing	Capturing video right now	None — normal
White	Steady	Booting up	Wait 2-3 minutes
Cyan	Steady	Maintenance mode	Intentional — see Section 5
Green	Slow pulse	Shutting down	Normal (Sukabumi between cycles)

LED Color	Pattern	Meaning	Action
Red	Steady	Camera offline	May be normal (see note below)
Red	Flashing	Capture failed	See Section 7
Blue	Steady/flash	Network/LTE problem	See Section 7
Yellow	Steady/flash	Storage problem	See Section 7
Magenta	Steady	Power problem	See Section 7
OFF	—	No power or sleeping	Normal for Sukabumi between cycles

Note on solid red: A steady red LED means the camera is offline. On Sukabumi this is **normal between capture cycles** — the camera powers down with the PoE relay to save solar energy. Solid red is only a problem if the camera should be on at that time (e.g., Jakarta where the camera runs 24/7, or Sukabumi during an active capture cycle).

Rule of thumb: Green = good. Any other color for more than 10 minutes = investigate.

---

2. Normal Operation

Sukabumi (Solar)

The station wakes every 15 minutes, captures a video, processes it, uploads results, then goes back to sleep. A complete cycle takes about 3 minutes.

What you'll see: 1. LED turns white (booting) — ~25 seconds 2. LED turns green flashing (capturing) — ~1 minute 3. LED turns green steady (processing/uploading) — ~1-2 minutes 4. LED turns off (sleeping) — ~12 minutes 5. Repeat

Between cycles, the station is OFF. This is normal — it saves solar power.

Jakarta (AC Power)

The station runs continuously. It captures a video every 15 minutes but stays powered on between captures.

What you'll see: - LED is green steady most of the time - LED flashes green briefly every 15 minutes during capture - Station reboots itself once per day (health check) — you'll see white LED briefly

---

3. Checking the Station Remotely

Remote Access (Pangolin)

The primary way to access the station dashboard from anywhere with internet:

- **Jakarta:** `https://arc-00001.openrivercam.com`
- **Sukabumi:** `https://arc-00002.openrivercam.com`

This provides HTTPS access to the ORC-OS web dashboard via a tunneled reverse proxy. No port number needed — just the URL above. Login password is stored separately (ask Tom).

The dashboard shows recent videos, processing status, and sensor data.

Note: We are also evaluating Tailscale as an alternative remote access solution. Tailscale may be a better fit in some situations but is not usable in countries that restrict third-party VPN services.

LiveORC Server

All processed data uploads to: `https://openrivercam.endlessprojects.info/`

Login to see discharge measurements, time series, and video thumbnails from all stations.

---

4. Power Button

Each station has a single waterproof button on the enclosure.

Action	What happens
Brief press (Pi is off)	Powers on the station
Brief press (Pi is running)	Clean shutdown
Long press (3 seconds)	Enters maintenance mode
Hold 10 seconds	Forces power off (emergency only)

Sukabumi note: You normally do NOT need the power button. The Witty Pi schedule handles wake/sleep automatically. Only use the button for maintenance.

---

5. Power Cycle — How the Station Wakes and Sleeps

Each station has a **Witty Pi 5 HAT+** board that controls when the Pi receives power. The Witty Pi has its own clock (CR2032 battery) and runs independently of the Pi — even when the Pi is completely off, the Witty Pi is keeping time and waiting for the next scheduled wake.

How It Works (Sukabumi — Duty Cycled)

The power cycle has two parts with **split responsibility**:

1. **Witty Pi turns the Pi ON** at the scheduled time (restores 5V power)
2. The Pi boots, captures video, processes it, uploads results
3. **ORC-OS shuts the Pi DOWN** when processing is complete ("shutdown after task")
4. The Pi is completely off — no power draw except the Witty Pi clock
5. **Witty Pi turns the Pi ON again** at the next scheduled time

The Witty Pi's ON window (e.g., 10 minutes) is a **safety backstop** — if ORC-OS doesn't shut down on its own (software hang, crash), the Witty Pi forces a clean shutdown at the end of the ON window. Under normal operation, ORC-OS shuts down well before this backstop kicks in.

How It Works (Jakarta — Always On)

Jakarta's Witty Pi is set to "default state when powered = ON." The Pi stays on continuously. The Witty Pi just passes power through. ORC-OS reboots the Pi once per day as a health check.

Available Schedules (Sukabumi)

Schedule	ON	OFF	Captures/day	Use when
prod_15 (default)	10 min	5 min	96	Normal operation
prod_30	25 min	5 min	48	Low solar / conserve battery
maint	Always on	—	Continuous	Debugging, software updates

Changing the Schedule

This requires SSH access to the Pi via Pangolin or Tailscale.

1. SSH into the station
2. Run `wp5` (the Witty Pi interactive menu)
3. Select option **6** to choose a schedule file
4. Select the desired `.wpi` schedule
5. The new schedule takes effect on the next power cycle

To check the current schedule:

```
wp5
```

Option **7** shows the current schedule status.

To switch Jakarta to duty-cycle mode (e.g., if moved to solar power in the future): load a `.wpi` schedule file and set ORC-OS "Shutdown after task" to ON.

To switch Sukabumi to always-on (for extended maintenance): load the `maint.wpi` schedule. Remember to switch back to `prod_15.wpi` when done.

---

6. Maintenance Mode

Maintenance mode **suspends video capture**, which also suspends the ORC-OS "shutdown after task" behavior. This is useful for debugging, software updates, or any work where you don't want the station capturing or shutting down mid-task.

> **Important:** Maintenance mode does NOT change the Witty Pi schedule. On Sukabumi, the Witty Pi will still cut power at the end of its ON window (e.g., after 10 minutes) even in maintenance mode. If you need the station to stay on longer than one Witty Pi cycle, you must also load the `maint.wpi` schedule (see Section 5).

Entering Maintenance Mode

Option A — Power button: Long press (3 seconds). LED turns cyan.

Option B — Remote (GitHub): Someone with repository access changes the station's mode file from "production" to "maintenance" at:

`https://github.com/tom-jordan23/orc-pmi-stations`

Use the "Set Station Mode" workflow (Actions tab) to toggle a station between production and maintenance. The station picks up the change on next boot.

What Changes in Maintenance Mode

- Video capture stops (no RTSP pulls, no relay cycling)
- ORC-OS "shutdown after task" is effectively disabled (nothing to trigger it)
- ORC-OS web dashboard stays accessible
- LED shows cyan
- **Witty Pi schedule is unchanged** — the Pi will still power off at the end of the ON window unless you load `maint.wpi`

For Extended Maintenance on Sukabumi

To keep the station on indefinitely: 1. Enter maintenance mode (button or GitHub) 2. SSH in via Pangolin and load the maintenance schedule: `wp5` → option 6 → `maint.wpi` 3. The Witty Pi will now keep the Pi powered continuously 4. When done: load `prod_15.wpi`, set station back to "production", reboot

Exiting Maintenance Mode

- Change mode back to "production" on GitHub and reboot, OR
- Reboot the Pi (`sudo reboot` via SSH)
- On Sukabumi: also restore the production schedule (`prod_15.wpi`) if you changed it

---

7. Routine Maintenance

Monthly

- ☐ **Visual check:** LED is green? Enclosure intact? No water intrusion?
- ☐ **Camera lens:** Clean with microfiber cloth if dirty (insects, dust, rain spots)
- ☐ **Rain gauge dome:** Wipe clean if debris has accumulated (leaves, bird droppings)
- ☐ **Cable glands:** Visually inspect — no cracking, no loose fittings
- ☐ **Check LiveORC:** Log in and verify recent data is uploading

Quarterly

- ☐ **Open enclosure** — read the warning below first:

  WARNING: Opening the enclosure in humid conditions can cause damage. The enclosure uses GORE vents to equalize pressure while keeping water out. The air inside the box is drier than the outside air. When you open the lid, humid tropical air rushes in. If the electronics are cooler than the incoming air (e.g., after a cool night or rain), moisture will **condense on the circuit boards** — this can cause shorts and corrosion.

  Rules for opening the enclosure:

  - **Do not open if ambient humidity is above 70% RH** (use a weather app or the SHT40 reading from the dashboard to check)

  - **Never** open during or immediately after rain
  - Open during the **warmest, driest** part of the day (midday sun)
  - Take your time once open — the air exchange happens immediately when the lid opens, so rushing won't help
  - If you see condensation on any component, close the lid immediately and let the GORE vents do their job over the next 24 hours
  - Enter maintenance mode before opening (prevents capture cycles during service)

  Once open, check:

  - Moisture, insects, corrosion inside
  - Terminal blocks — no loose wires
  - Fuses — no discoloration
  - LED and sensors are clean

- ☐ **Sukabumi solar:** Clean solar panel if dirty. Check battery voltage (should be >12V). Inspect cable connections at battery terminals.

After Storms

- ☐ Check station is still operating (LED green)
- ☐ Check camera is still aimed correctly (hasn't shifted)
- ☐ Check rain gauge hasn't been knocked off mount
- ☐ Check enclosure for water intrusion
- ☐ **Jakarta:** Check AC power is on (building may have tripped breaker)

---

8. Troubleshooting — Quick Fixes

Station LED is OFF (no light at all)

1. **Sukabumi:** May be sleeping between cycles. Wait 15 minutes — if LED doesn't come on, press power button briefly.
2. **Jakarta:** Check AC power is on at the building breaker.
3. Press power button briefly. If nothing happens, open enclosure and check fuses (see door sheet inside lid).

LED is RED (camera problem)

1. Camera may need time to boot. Wait 5 minutes.
2. Check that the ethernet cable between enclosure and camera is connected.
3. Open enclosure → check the PoE switch has power (green LED on the switch).
4. If PoE switch is dark, check fuse F2 (5A).

LED is BLUE (network problem)

1. LTE modem may need time to connect. Wait 5 minutes.
2. Check antenna — is it connected and not damaged?
3. If problem persists, check SIM card (open enclosure, inspect modem).
4. Verify cell coverage at site hasn't changed.

LED is YELLOW (storage problem)

1. Storage may be full. Connect to web dashboard and check disk space.
2. If disk is full, old videos should auto-clean. If not, contact Tom.

LED is MAGENTA (power problem)

1. **Sukabumi:** Battery may be low. Check if solar panel is clean and unobstructed. Check battery voltage (should be >12V).
2. **Jakarta:** Check AC power. May be a brownout or voltage sag.

No data on LiveORC server

1. Station may be working but LTE upload is slow. Wait 1 hour.
2. Check station LED — if green, station is healthy. Problem is connectivity.
3. Check if SIM card has data remaining (prepaid plans expire).
4. Remote-access the station via Pangolin and check upload status.

Camera image is dark or blurry

1. Clean the camera lens (microfiber cloth).
2. Check that nothing is blocking the camera view (new vegetation, spider webs).
3. IR LEDs handle night vision automatically — no adjustment needed.

---

9. What NOT to Do

Do NOT	Why
Disconnect wires inside the enclosure	You may connect them wrong and damage the Pi
Change any software settings without guidance	The system is carefully configured
Open the enclosure in rain	Water intrusion damages electronics
Force-hold the power button regularly	Use brief press for clean shutdown
Unplug the rain gauge cable	It accumulates data that would be lost
Point the camera at a different angle	Requires complete recalibration (field survey)
Swap SD cards between stations	Each card is configured for its station
Connect power to any terminal marked “GPIO”	12V on GPIO pins destroys the Pi instantly

---

10. Sensor Reference

Rain Gauge (Hydreon RG-15)

- Mounted outside the enclosure

- Optical sensor — no moving parts, self-cleaning dome
- Measures rainfall intensity and accumulation
- Stays powered 24/7 (even when Pi sleeps on Sukabumi)
- **Cleaning:** Wipe dome with damp cloth. Do not use solvents.
- **Disconnect:** SD16 4-pin connector (keyed, waterproof). Unplug to replace gauge without opening enclosure.

Figure 1: Hydreon RG-15 optical rain gauge showing clear dome and PCB

Figure: The RG-15 rain gauge. The clear dome is the sensing surface — keep it clean. No moving parts inside.

Temperature/Humidity (SHT40)

- Mounted inside the enclosure
- Monitors internal conditions (humidity alert threshold)
- No maintenance required

Outside Temperature (DS18B20)

- Waterproof probe mounted outside the enclosure
- Passes through PG9 cable gland
- No maintenance required

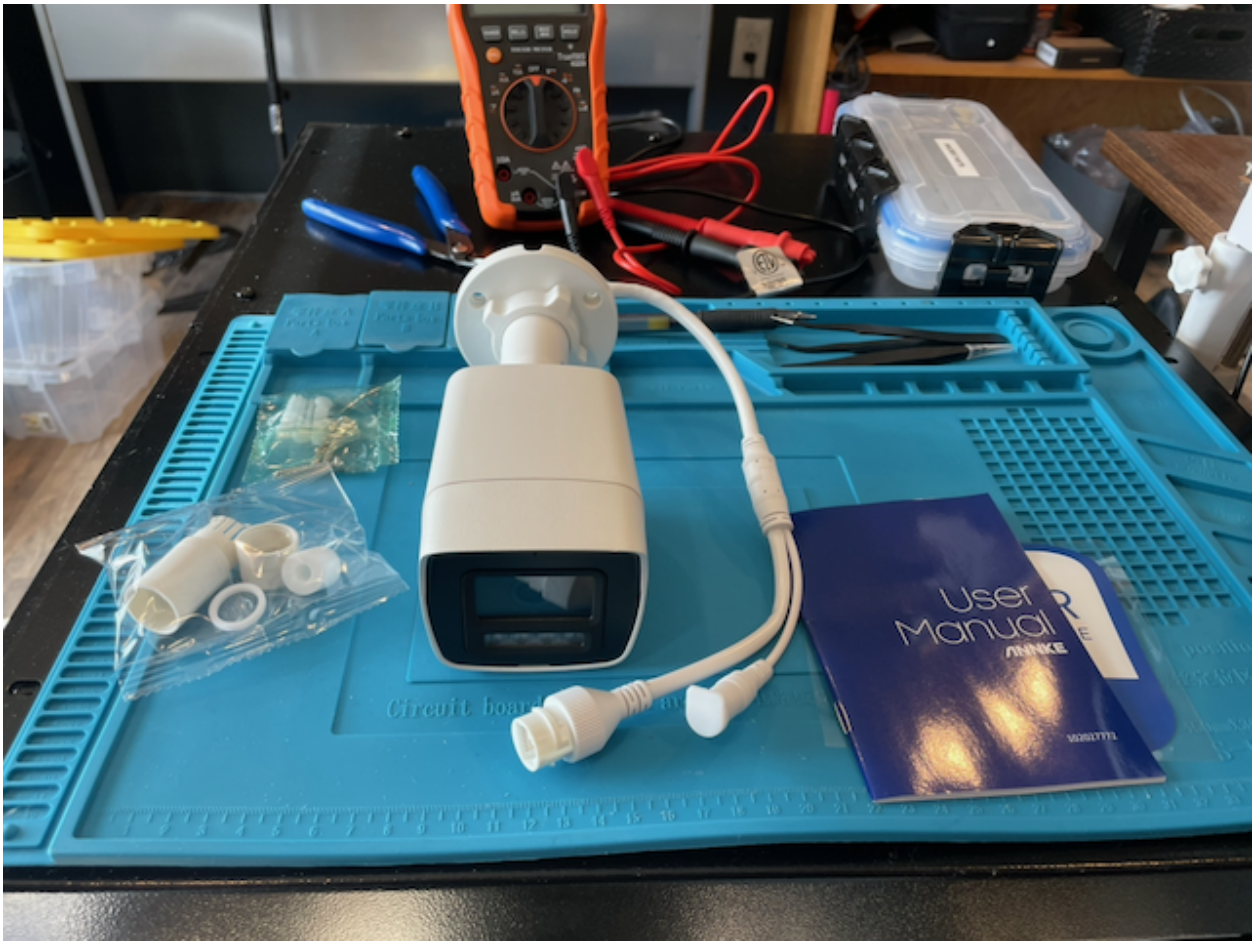

Figure 2: ANNKE C1200 PoE camera with mounting hardware

Figure: The ANNKE C1200 PoE camera installed at each site. Factory-sealed IP67 housing with built-in IR LEDs for night operation.

---

11. Extending the Station — Available Relay Channels

Each station has a 4-channel relay module. Only **channel 1** is used (PoE camera power). **Channels 2, 3, and 4 are available** for future use.

What the Relays Can Do

Each relay channel is an electrically isolated switch that the Pi controls via a GPIO pin. When the Pi sets the pin HIGH, the relay closes and connects power to whatever is wired to it. When the pin goes LOW (or the Pi loses power), the relay opens and the load loses power.

- **Switching capacity:** 12V DC from the station's power bus (TB1)
- **Each channel is independent** — they can be on/off in any combination
- **Fail-safe:** All relays open (loads off) if the Pi crashes or loses power
- **Control:** Any script or service on the Pi can toggle a relay by setting a GPIO pin high or low

Ideas for Future Use

Alerting and notification: - Wire a relay to a siren, strobe light, or warning beacon. Trigger it from software when the river level exceeds a threshold. The relay can power a 12V alarm device directly from TB1. - Wire a relay to a cellular SMS gateway module (12V powered). Send automated text messages during flood events.

Additional instruments: - Power a water level sensor on a duty cycle (turn on to take a reading, turn off to save power). Useful for Sukabumi where solar budget is limited. - Power a second camera for a different view angle or a wider scene. - Power a 12V solenoid for automated water sampling.

Integration with other systems: - Use a relay as a dry-contact output to interface with existing telemetry or SCADA systems. The relay's NO/NC terminals can signal to any system that reads contact closures. - Power a 12V radio or LoRa transmitter for local data relay to a nearby station or gateway.

How to Connect a New Load

This requires opening the enclosure. Only trained personnel should do this. See Section 6 for humidity warnings about opening the enclosure.

The GPIO control wiring from the Pi to relay inputs IN2, IN3, and IN4 is **already done** on both stations. You only need to wire the 12V load side.

Each unused relay channel has three screw terminals on the 12V output side:

Terminal	Function
COM (Common)	Connect to 12V+ from TB1 (through a fuse)
NO (Normally Open)	Connect to your 12V load (+)
NC (Normally Closed)	Not used (leave empty for fail-safe behavior)

The load's ground wire returns to TB1 GND.

Always add a fuse between TB1 and the relay COM terminal. Match the fuse to the load's current draw (e.g., 2A fuse for a 20W load at 12V).

To control the relay from the Pi (command line):

```
# Turn on relay channel 2 (GPIO 17)
gpioset gpiochip0 17=1

# Turn off
gpioset gpiochip0 17=0
```

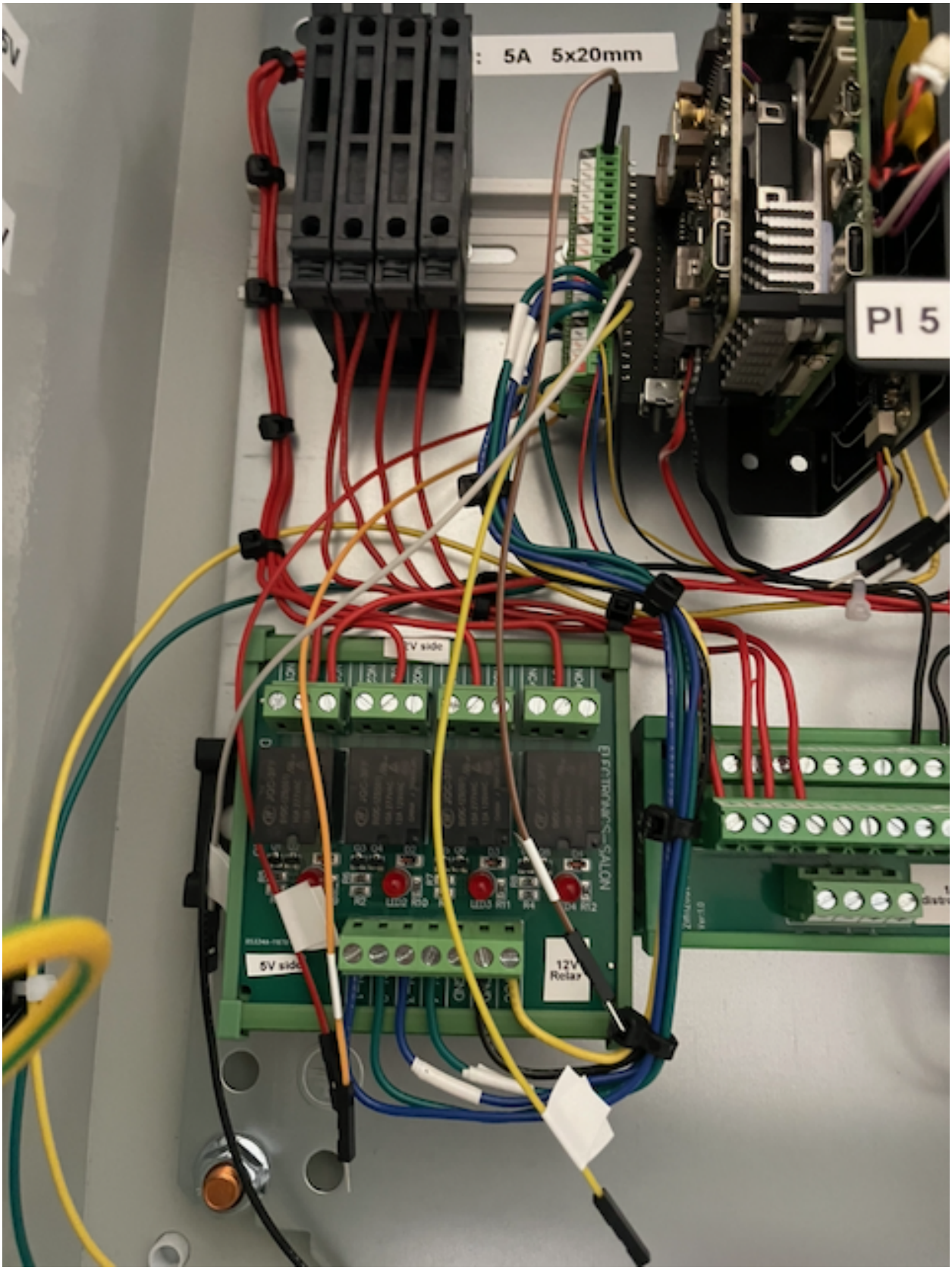

Figure 3: Inside the enclosure showing DIN rail fuse holders, relay module with green PCBs, and G469 GPIO terminal connections

Figure: Inside the enclosure. The relay module (green PCBs, center) has four channels — CH1 powers the camera, CH2-4 are available. Each has COM/NO/NC screw terminals on the 12V side.

Relay Channel Assignments

Channel	GPIO	G469 Pin	Status
CH1	GPIO 24	Pin 18	**In use** — PoE camera power
CH2	GPIO 17	Pin 11	Available — GPIO wired, no load
CH3	GPIO 27	Pin 13	Available — GPIO wired, no load
CH4	GPIO 22	Pin 15	Available — GPIO wired, no load

These assignments are the same on both stations.

Important Notes

- **Do not exceed 12V or 10A per channel** on the relay contacts
- **Always use NO (Normally Open)** contacts for fail-safe behavior — loads turn off if the Pi loses power
- **Add a fuse** on every new load circuit
- **Label all new wiring** clearly at both ends
- **Document what you connected** — update the door sheet inside the enclosure
- **Test before leaving the site** — verify the relay toggles and the load responds

---

12. Emergency Contacts

Role	Name	Phone	Email
Technical lead	Tom Jordan	*(fill in)*	*(fill in)*
PMI site contact (Sukabumi)			
PMI site contact (Jakarta)			

If the station is completely unresponsive:

1. Note the LED color (or that it's off) and the time
2. Take a photo of the station and any visible damage
3. Contact Tom with the photo and description
4. **Do not open the enclosure or disconnect anything** unless instructed